

Engineering Mechanics

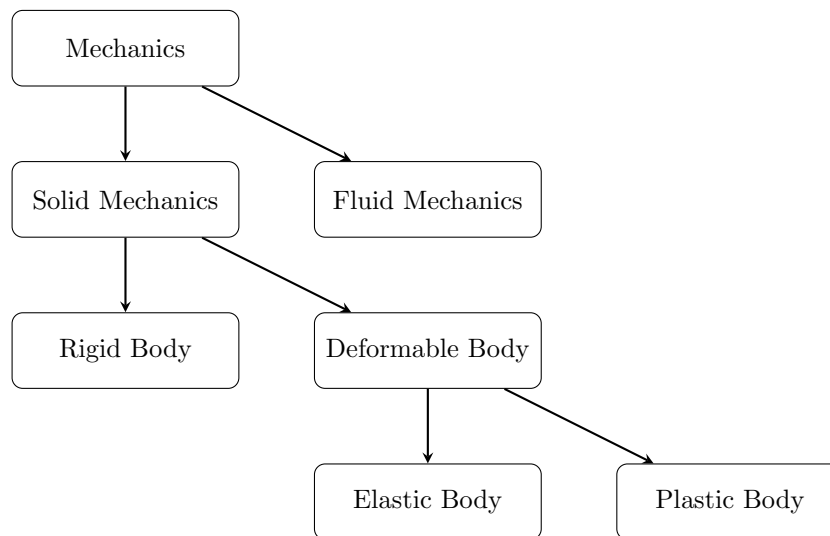
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1 Statics

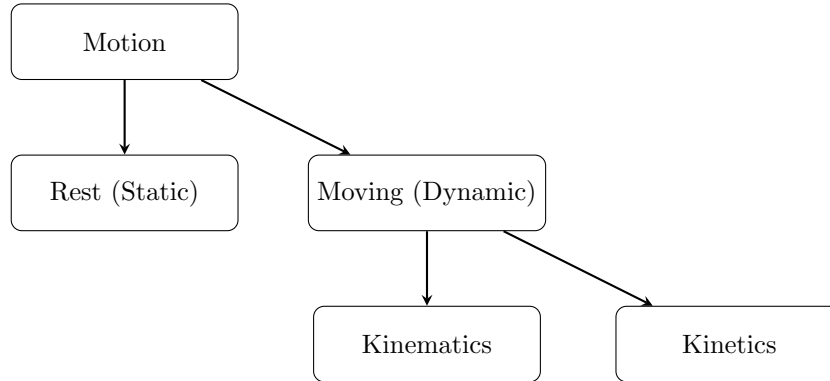
1.1 Mechanics

The study of the **motion** (or rest) a **body** under the effect of an **external agent** is called as **Mechanics**



From all these, we need to only study about **Rigid Body Mechanics**.

1.2 Types of motion



1.3 Idealization of Mechanics

1. **Continuum:** The effect of force applied on a body is on the entire body.
2. **Rigid Body:** Deformation of the body is negligible.
3. **Concentrated Force:** Force applied on the body is concentrated at one point on the body, usually the center of mass.
4. **System of Particles:** A particle is a mass with negligible dimension.

1.4 Fundamental Laws of Mechanics

1. **Newton's 1st Law:** A body at rest or in uniform motion remains in that state unless an external and unbalanced force is acted on it.
2. **Newton's 2nd Law:** The rate of change of linear momentum is directly proportional to the force and is in the direction of the force. $\frac{d\vec{p}}{dt} = \vec{F} = m\vec{a}$
3. **Newton's 3rd Law:** Every action has an equal and opposite reaction.
4. **Newton's Law of Gravitation:** Every body in the universe attracts every other body with a force directly proportional to the product of the two masses and inversely proportional to the square of distance between the two. $F = \frac{Gm_1m_2}{d^2}$
5. **Law of Transmissibility:** The state of rest or motion of a rigid body is unaltered if force acting on the body is replaced by another force of same magnitude and direction, but anywhere along the line of force.
6. **Law of parallelogram of vector addition:** Two forces acting on a particle may be represented by a single force, i.e. the resultant obtained by drawing the diagonal of the parallelogram which has sides equal to the given force.

$$|\vec{R}| = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$

$$\tan\alpha = \frac{F_2\cos\theta}{F_1 + F_2\sin\theta}$$

Triangle Law: If two forces are acting on a body are represented one after another by sides of a triangle then the resultant is represented by the closing side of the triangle taken from the first point to the last one.

1.5 Types of Forces

1. **Collinear Force System:** System of forces in which lines of action of all forces are along the same line.
2. **Coplanar Parallel Forces:** System of forces in which all forces lie in the same plane, parallel to each other.
3. **Coplanar Like Parallel Forces:** All forces lie in the same plane, parallel to each other, in the same direction.
4. **Coplanar Concurrent Forces:** Line of action of all forces pass through a single point and all forces lie in the same plane.
5. **Non-coplanar Concurrent Forces:** Line of action of all forces meet at a single point, but not lie in the same plane.
6. **Non-coplanar Non-concurrent Forces**
7. **Coplanar Non-concurrent Forces**

1.5.1 Equilibrium

A particle is said to be in **equilibrium** when the resultant of all the forces acting on it is zero.

1.6 Moment of Force (Torque)

1. Torque, $\tau = F \cdot d$
2. Point above which force is acting is called **moment center** and the perpendicular distance from line of action to moment center is called **moment arm**.
3. **Varignon's Theorem:** Moment of the resultant of several concurrent forces about a given point 'O' is equal to the sum of moments of individual forces about 'O'.

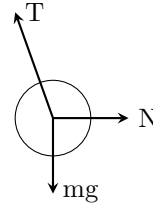
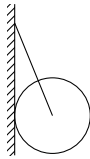
Example: $\tau_1 = rF_1, \tau_2 = rF_2, \tau_3 = rF_3 \implies \tau_{net} = r(F_1 + F_2 + F_3)$

1.7 Couple:

1. Two parallel forces equal in magnitude, opposite in direction, and separated by a finite distance is called a **couple**.
2. Moment due to couple is only dependent on distance by which they are separated. Means, $\tau_{anywhere} = rF$
3. Translational effect of couple forces in any direction is zero.
4. The effect of couple remains the same even if it is replaced by another set of couple forces having the same moment. (Example: 10N forces separated by 2m **and** 5N forces separated by 4m).
5. Any concentrated moment loading can be represented as a couple.

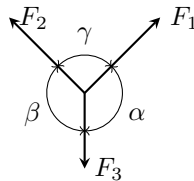
1.8 Free Body Diagrams

1. It is a diagram of the body in which the body is free from all the contact surfaces, and is shown with all the forces on it.
2. Example:



1.9 Lami's Theorem

If a body is in equilibrium under the action of only three forces, then each force is proportional to the sine of the angle between the other two forces.



$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

1.10 Structures

There are three types of structures:

1. **Truss:** Fully constrained structures which are designed to support loads. Every member in a truss is a 2-force member.
2. **Frame:** Same as a truss, but has at least one multi-force member.
3. **Machines**

1.10.1 Simple / Plane Truss:

1. A plane truss is a truss consisting of members which lie in a plane and are loaded in the same plane.
2. If n is the number of members, and j is the number of joints, then:
 - (a) If $n = 2j + 3$, the truss is said to be **perfect**.
 - (b) If $n < 2j + 3$, the truss is said to be **redundant**.
 - (c) If $n > 2j + 3$, the truss is said to be **deficient**.
3. **Truss Analysis - Method of Joints:**
 - (a) Draw the free body diagram of the structure.
 - (b) Apply equilibrium conditions considering whole structure as one.
($\sum F_x = 0$, $\sum F_y = 0$, $\sum \tau = 0$)
 - (c) Assuming members are in tension (internal forces are away from the joints), find the reaction at all the joints.
4. **Truss Analysis - Method of Sections:**
 - (a) Draw the free body diagram of the structure.
 - (b) Apply the equilibrium conditions for the entire structure as one.
 - (c) Select a section of the structure abiding the following rules:
 - i. It must pass through a maximum of 3 members.
 - ii. It must pass through the member on which force is to be found.
 - iii. It must not pass through joints.
 - (d) Apply $\sum F_x = 0$, $\sum F_y = 0$, $\sum \tau = 0$ for the section.

1.11 Distributed Load

If forces are applied over a region whose dimension is not negligible as compared to other body/structure dimension, then the forces applied are distributed.

1. **Center of Mass (CM):** A point where the entire mass of a body is assumed to be concentrated.
2. **Center of Gravity (CG):** The point along where the gravitation force on a body is being acted. If g is constant, then $CG=CM$.
3. **Centroid:** The geometrical center of a 2-D object.

Consider a rigid body whose CM is y units above x -axis and x units in front of y -axis. Also, assume the weight of the body be $W = \int dW$

Using Varignon's Theorem, $Wx = \int dWx_i$

$$x = \frac{\int_0^i x_i dW}{W} \implies y = \frac{\int_0^i y_i dW}{W}$$

If mass is m and g is constant ($CM=CG$),

$$x = \frac{\int_0^i x_i dm}{m} \implies y = \frac{\int_0^i y_i dm}{m}$$

If volume is V and ρ is constant,

$$x = \frac{\int_0^i x_i dV}{V} \implies y = \frac{\int_0^i y_i dV}{V}$$

1.11.1 Centroid of a Rectangle (b, h):

Consider a rectangle of height h and breadth b . Take a small elemental strip of width dx at a distance of x from the y -axis (assuming the side of rectangle coincides with the y -axis). Then,

$$\bar{x} = \frac{\int x dA}{A} = \frac{\int_0^b x(hdx)}{bh} = \frac{hx^2}{2bh} \Big|_0^b \implies \bar{x} = \frac{b}{2}$$

Similarly, $\bar{y} = h/2$

Hence, location of centroid is: $(b/2, h/2)$.

1.11.2 Centroid of an Isosceles* Triangle (h, b):

* Here Isosceles means symmetric to y-axis.

Consider a triangle of base b and height h. Take a small elemental strip of height dy at a distance of y from the x-axis. Correspondingly, let the width of strip be x.

$$\bar{y} = \frac{\int y dA}{A} = \frac{\int_0^h y(x dy)}{\frac{1}{2}bh}$$

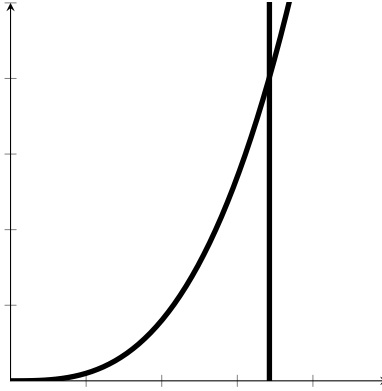
According to c.s.c.t,

$$\begin{aligned} \frac{h}{b} &= \frac{h-y}{x} \implies x = \frac{b(h-y)}{h} \\ \therefore \bar{y} &= \frac{2}{bh} \int_0^h \frac{b}{h} (h-y)y dy = \frac{2}{h^2} \int_0^h (hy - y^2) dy \\ \implies \bar{y} &= \frac{2}{h^2} \left[\frac{hy^2}{2} - \frac{y^3}{3} \right]_0^h = \frac{2}{h^2} \left[\frac{h^3}{2} - \frac{h^3}{3} \right] \implies \bar{y} = \frac{h}{3} \end{aligned}$$

Also, by symmetry, $\bar{x} = b/2$

Hence, location of centroid is (b/2, h/3).

1.11.3 Centroid of curve $y = kx^3$:



Assume a vertical strip of width dx is taken from a distance of y from the x-axis and x from y-axis.

$$dA = y dx \implies A = \int_0^a y dx = \int_0^a kx^3 dx \implies A = \frac{1}{4}ka^4$$

$$\text{Now, } (a,b) \implies b = ka^3 \implies k = ba^{-3} \implies A = \frac{1}{4}(ba^{-3})a^4 \implies \boxed{A = \frac{1}{4}ab}$$

$$\bar{x} = \frac{\int_0^a x dA}{A} = \frac{\int_0^a x(kx^3 dx)}{A} = \frac{k}{A} \int_0^a x^4 dA = \frac{k}{5A} x^5 \Big|_0^a$$

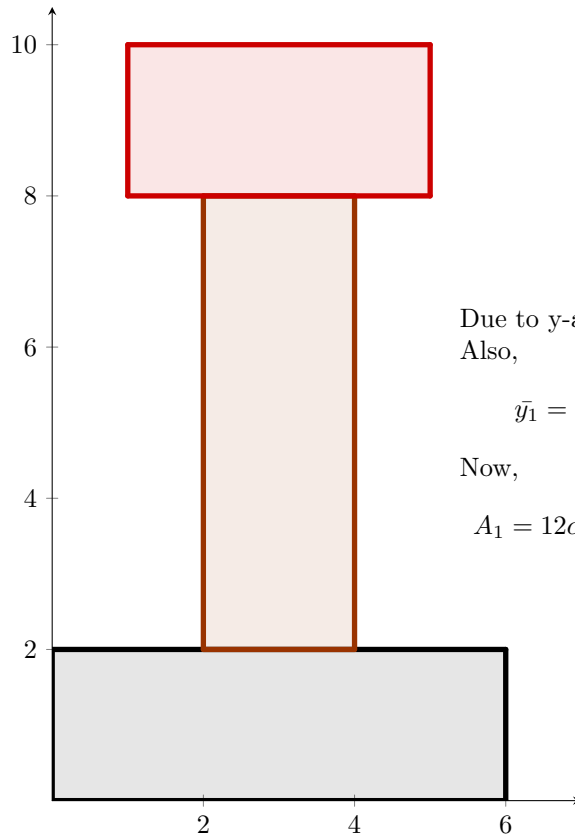
$$\Rightarrow \bar{x} = \frac{ba^{-3}}{5(\frac{1}{4}ab)}(a^5) = \frac{4ba^5}{5a^4b} \Rightarrow \boxed{\bar{x} = \frac{4}{5}a}$$

Now, assume a horizontal strip of height dy is taken from a distance of x from the y -axis and y from x -axis. Also, $x = \sqrt[3]{\frac{y}{k}}$

$$y = \frac{1}{A} \int_0^b y \left(\frac{y^{1/3}}{k^{1/3}} \right) dy = \frac{1}{k^{1/3}A} \int_0^b y^{4/3} dy = \frac{7}{3k^{1/3}A} y^{7/3} \Big|_0^b$$

$$\Rightarrow \bar{y} = \frac{7}{3(b/a^3)^{1/3}(ab/4)}(b^{7/3}) = \frac{28ab^{4/3}}{3b^{1/3}ab} \Rightarrow \boxed{\bar{y} = \frac{28}{3}b}$$

1.11.4 Centroid of a multi-shape figure:



Due to y -axis symmetry, $\bar{x} = 3cm$
Also,

$$\bar{y}_1 = 1cm, \bar{y}_2 = 5cm, \bar{y}_3 = 9cm$$

Now,

$$A_1 = 12cm^2, A_2 = 12cm^2, A_3 = 8cm^2$$

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i} = \frac{12 + 60 + 72}{12 + 12 + 8} = 4.5cm$$

1.12 Pappus-Guldinus Theorems:

1. The area of a surface of revolution is the product of the length of generating curve and the distance travelled by the centroid of the curve while the surface is being generated.
2. The volume of a body of revolution is obtained from the product of the generating area and the distance traveled by the centroid of the area while the body is being generated.

1.13 Moment of Inertia (MI):

1. **Plane/Mass Moment:** $I = MR^2$. This is called second-order moment.
2. $I = (MR)R$ and $MR =$ Centroid Moment, called a first-order moment.
3. Mass MI $= \int r^2 dm$
4. Area MI $= \int r^2 dA$
5. MI about X axis, $I_x = \int y^2 dA$
6. MI about Y axis, $I_y = \int x^2 dA$
7. **Perpendicular Axis Theorem:** The MI of a plane area about the axis perpendicular to the plane and passing through the intersection of other two axes XX' and YY' contained by the plane is equal to the sum of MI about XX' and YY'. Hence,

$$I_z = I_x + I_y$$

8. **Radius of gyration:** The radial distance to a point which would have a moment of inertia the same as the body's actual distribution of mass, if the total mass of the body were concentrated there.

$$I = k^2 A \implies k = \sqrt{\frac{I}{A}}$$

9. **Parallel Axis Theorem:** The moment of inertia about any axis parallel to the centroidal axis is equal to the sum of the moment of inertia about the centroidal axis and the product of the area and square of the perpendicular distance between the axes.

$$I_a = I_c + ah^2$$

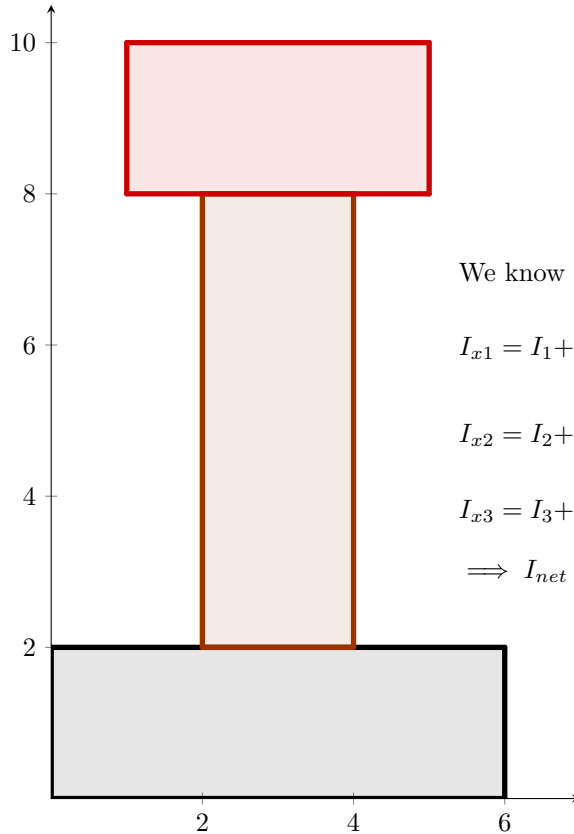
1.13.1 Moment of Inertia about centroidal axis of simple figure:

Consider a rectangle of width b and height h . Its centroidal axis will be at a height of $h/2$ from the top. Let us take an elemental strip of area dA and height dy from the centroidal axis at a distance of y .

$$I_x = \int dI_x = \int y^2 dA = b \int_{-h/2}^{h/2} y^2 dy = \frac{1}{3} b y^3 \Big|_{-h/2}^{h/2} \Rightarrow I_x = \frac{1}{12} b h^3$$

Similarly, $I_y = \frac{1}{12} b^3 h$

1.13.2 Moment of Inertia about centroidal axis of multi-shape figure:



We know that $\bar{x} = 3cm$ and $\bar{y} = 4.5cm$

$$I_{x1} = I_1 + a_1 h_1^2 = \frac{(6)(2)^3}{12} + (12)(3.5)^2 = 151cm^4$$

$$I_{x2} = I_2 + a_2 h_2^2 = \frac{(2)(6)^3}{12} + (12)(0.5)^2 = 39cm^4$$

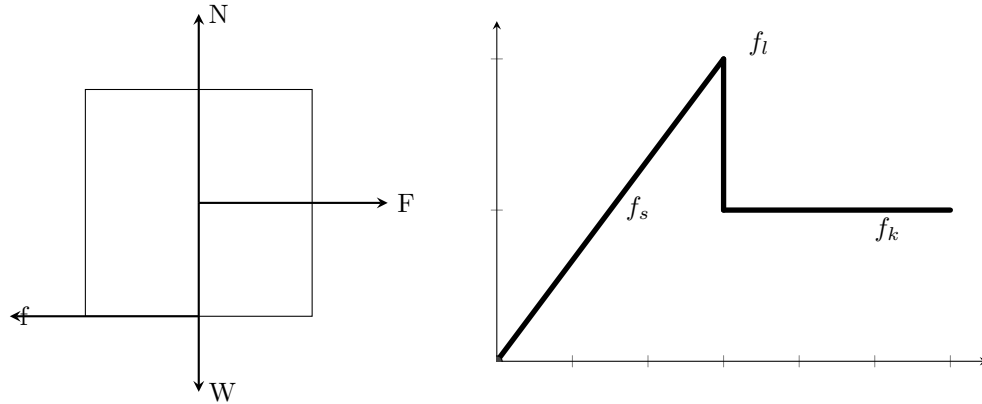
$$I_{x3} = I_3 + a_3 h_3^2 = \frac{(4)(3)^3}{12} + (8)(4.5)^2 = 164.67cm^4$$

$$\Rightarrow I_{net} = 151 + 39 + 164.67 = 354.67cm^4$$

1.14 Friction

1. Friction is the force opposing relative motion.
2. **Types of Friction:**
 - (a) External Friction
 - i. **Dry Friction:** Friction developed between unlubricated surfaces
 - ii. **Fluid Friction:** Friction developed between two layers of a fluid
 - (b) Internal Friction
3. **Impending Motion:** Motion is said to be "impending" if applied force is such that the body is just about to slide.

1.14.1 Dry Friction:



Coefficient of static friction, $\mu_s = \frac{f_s}{N}$
 Coefficient of kinetic friction, $\mu_k = \frac{f_k}{N}$

$$\tan\phi_s = \frac{f_l}{N} = \frac{\mu_s N}{N} \Rightarrow \boxed{\tan\phi_s = \mu_s} \Rightarrow \boxed{\tan\phi_k = \mu_k}$$

Characteristics of Dry Friction:

1. The frictional force acts tangential to surface of contact and in direction opposite to the relative / impending motion of one surface against another.
2. The magnitude of maximum static frictional force that can be developed is independent of contact area provided that normal pressure is neither small nor large enough to deform or crush surfaces of contact.
3. The magnitude of maximum static frictional force is usually larger than that of kinetic frictional force for any two contact surfaces. $f_s \approx f_k$ when $\mu_s = \mu_k$ i.e. one body is moving with a very low velocity over another.
4. If body is just about to slide, $f_l \propto N \Rightarrow f_l = \mu_s N$
5. If slipping takes place, $f_k \propto N \Rightarrow f_k = \mu_k N$

1.15 Virtual Work:

1. A system of connected rigid bodies is in equilibrium provided that the virtual work done by all the external forces and couples on the system is zero for each independent virtual displacement of the system.
2. **Procedure:**
 - (a) Draw the FBD of the system.
 - (b) Specify the coordinates $\delta x, \delta y$
 - (c) Apply virtual work principle $\sum W = 0$

The below are taken from the PPTX/PDF Taken from
../References/Books/ME100

2 Dynamics

1. Dynamics is the study of an object under motion.
2. **Kinematics:** The study of geometry of motion without the reference to the cause of motion.
3. **Kinetics:** The study of relations between force, mass, motion of body.
4. In dynamics, a particle is assumed to have a mass, but its size and shape cannot be ignored.

2.1 Kinematics of Particle:

2.1.1 Rectilinear Motion:

1. Motion of an object in a straight line.
2. **Types:** Continuous motion, Erractic motion.

2.1.2 Curvilinear Motion:

1. Motion of an object in a curved path in 2-D or 3-D.
2. **Types:** Rectangular components, Normal-Tangent components, Cylindrical components.

2.1.3 Displacement, Velocity, Acceleration:

1. **Distance:** The total path length traversed by an object.
2. **Displacement:** The shortest path between an initial and a final position.
3. **Velocity:** The rate of change of displacement with respect to time.

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

4. **Acceleration:** The rate of change of velocity with respect to time.

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

Also,

$$a = v \frac{dv}{dx} \implies adx = vdv$$

2.2 Rectilinear Motion

2.2.1 Continuous Motion:

1.

$$\int_u^v dv = \int_0^t a dt \implies \boxed{v - u = at}$$

2.

$$\int_{x_0}^x dx = \int_0^t v dt \implies \boxed{x - x_0 = \Delta x = ut + \frac{1}{2}at^2}$$

3.

$$\int_u^v v dv = \int_{x_0}^x a dx \implies \boxed{v^2 - u^2 = 2a(x - x_0) = 2a\Delta x}$$

4. **Question:** A particle travels along a straight line to right with velocity $v = 4t - 3t^2$. Determine position, and acceleration of particle when $t = 4s$, given $x = 0$ when $t = 0$.

Answer:

$$a = \frac{dv}{dt} = 4 - 6t \implies a \Big|_{t=4s} = 4 - 6(4) = -20m/s^2$$

$$v = \int_0^4 v dt = [2t^2 - t^3] \Big|_0^4 = 2(16) - 64 = -32m$$

5. **Question:** A sandbag is dropped from a balloon which is ascending vertically at a constant speed of 6 m/s. If the bag is released with the same upward velocity of 6 m/s when $t = 0$ and hits the ground when $t = 8$ s, determine the speed of the bag as it hits the ground and the altitude of the balloon at this instant.

Answer: Displacement of balloon, $x_b = ut = (6)(8) = 48m$

Displacement of sandbag,

$$x_s = ut - \frac{1}{2}gt^2 = (6)(8) - \frac{1}{2}gt(9.8)(8)^2 = -265.9m$$

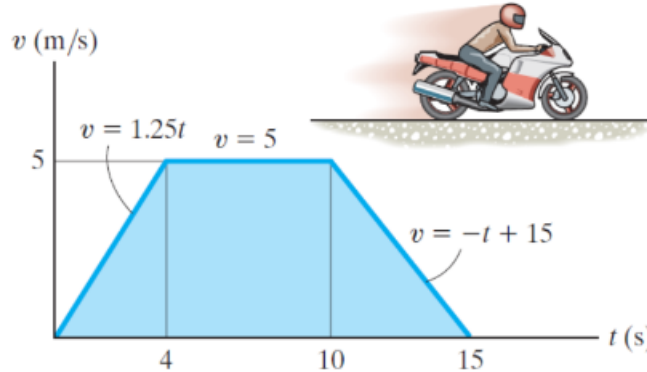
Hence, altitude of balloon, $x = x_b + x_s = 48 + 265.9 = 313.9m$

Now, velocity of bag, $v = u + at = (6) - (9.8)(8) = -74m/s$

2.2.2 Erratic Motion:

1. **x-t graph:** The displacement-time graph is used to find velocity by taking the slope of the curve.
2. **v-t graph:** The velocity-time graph is used to find displacement by taking the area under the curve and acceleration by taking the slope of the curve.
3. **a-t graph:** The acceleration-time graph is used to find velocity by taking the area under the curve.

4. **a-x graph:** $\frac{1}{2}(v^2 - u^2) = \int_{x_0}^x a dx$ = area under a-x graph.
5. **Question:** A motorcycle starts from rest at $s = 0$ and travels along a straight road with the speed shown by the v-t graph. Determine the motorcycle's acceleration and position over the time interval.



Answer: For $0 < t < 4$, $a = \frac{d}{dt}(1.25t) = 1.25 \text{ m/s}^2$
 For $4 < t < 10$, $a = \frac{d}{dt}(5) = 0$
 For $10 < t < 15$, $a = \frac{d}{dt}(-t + 15) = -1 \text{ m/s}^2$
 Also, $\Delta x = \frac{1}{2}(5)(6 + 15) = 52.5 \text{ m}$

2.3 Curvilinear Motion:

2.3.1 Rectangular Components:

1. In curvilinear motion, the direction of the instantaneous velocity is always tangent to the path.
2. In curvilinear motion, the direction of the instantaneous acceleration is always tangent to the hodograph.
3. A plot of the locus of points defined by the arrowhead of the velocity vector is called a hodograph. The acceleration vector is tangent to the hodograph, but not, in general, tangent to the path function.
4. It is often convenient to describe the motion of a particle in terms of its x , y , z or rectangular components, relative to a fixed frame of reference.
5. The position of the particle can be defined at any instant by the position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.
6. **Question:** A box slides down the slope described by the equation $y = (0.05x^2)$ m. $v_x = -3 \text{ m/s}$, $a_x = -1.5 \text{ m/s}^2$ at $x = 5 \text{ m}$. Find the y components of the velocity and the acceleration of the box at $x = 5 \text{ m}$.

Answer: $y = 0.05x^2 \implies v_y = 0.05(2x)(v_x) = 0.1(5)(-3) = -7.5m/s$
 $\implies a_y = 0.1(xa_x + v_x^2) = 0.1((5)(-1.5) + (-3)^2) = +0.15m/s^2$

7. **Question:** A particle travels along the path $y = 0.5x^2$. When $t = 0$, $x = y = z = 0$. Find the particle's distance and the magnitude of its acceleration when $t = 1$ s, if $v_x = (5t)$ m/s.

Answer: $y = 0.5x^2 \implies v_y = 0.5(2x)(v_x) = xv_x = 5xt$

$$x = \int_0^1 v_x dt = \int_0^1 5t dt = [2.5t^2]_0^1 = 2.5m \implies y = 0.5(2.5)^2 = 3.125m$$

Hence, $r = \sqrt{x^2 + y^2} = \sqrt{(2.5)^2 + (3.125)^2} \approx 4m$

Now, $a_y = 5[v_x t + x] = 5[(5) + (2.5)] = 5 \times 7.5 = 37.5m/s^2$

Hence, $a = \sqrt{x^2 + y^2} = \sqrt{(5)^2 + (37.5)^2} \approx 38.3m/s^2$

Hence, required values are: $r = 4m$, $a = 38.3m/s^2$.

2.3.2 Projectile Motion:

1. Projectile motion can be treated as two rectilinear motions, one in the horizontal direction (with zero acceleration) and the other in the vertical direction (with constant downward acceleration).

2. In projectile motion, velocity in the horizontal direction is constant.

3. Kinematic Equations in Horizontal Direction:

(a) $x = u_x t = u \cos \theta t$

(b) $v_x = u_x$

4. Kinematic Equations in Vertical Direction:

(a) $v_y = u_y - gt^2$

(b) $y = u_y t - 0.5gt^2$

(c) $v_y^2 = u_y^2 - 2gy$

5. Some Important Terms:

(a) Range of Projectile: $R = (u^2 \sin(2\theta))/g$

(b) Time of flight, $T = (2u \sin \theta)/g$

(c) Maximum height of the projectile, $H = (u^2 \sin^2(\theta))/2g$

6. **Question:** A ball is kicked with the initial velocity $v_A = 10m/s$ at an angle of 30° from the ground. Determine the range R, and the speed when the ball strikes the ground.

Answer:

$$R = \frac{u^2 \sin(2\theta)}{g} = \frac{100 \sin(60^\circ)}{9.81} = 8.83m$$

Now, $v_y^2 = u_y^2 - 2gy = u_y^2 - 2g(0) \implies v_y = -u_y = -u \sin \theta$

Also, $v_x = u_x = u \cos \theta$ Hence,

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(u \sin \theta)^2 + (u \cos \theta)^2} = u = 10m/s$$

7. **Question:** A projectile is fired with an initial velocity of $v_A = 150$ m/s off the roof of a building of height 150m at an angle of $\tan^{-1}(3/4)$. Determine the range R where it strikes the ground.

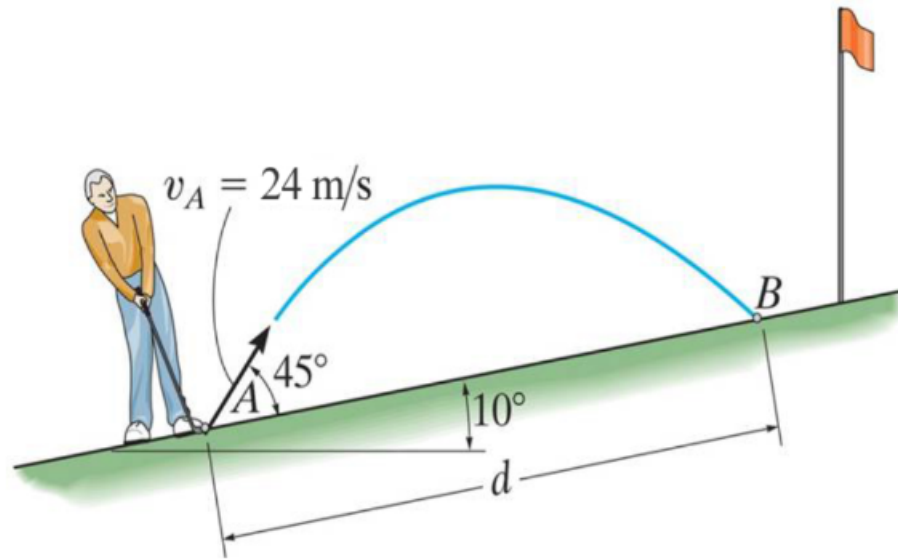
Answer:

$$x = u_x t \implies R = (v_A \cos \theta) t = 120t$$

$$y = u_y t - 0.5gt^2 = (v_A \sin \theta) t - 0.5(9.81)t^2 \implies -150 = 90t - 4.9t^2$$

$$\text{Solving for } t, \text{ we get } t = 19.89s \implies R = 120t = 2387m$$

8. **Question:** A golf ball is struck with a velocity of 24 m/s as shown. the distance d to where it will land.



Answer:

$$x = u_x t \implies d \cos(10) = 24 \cos(55) t \implies t = 0.071d$$

Now,

$$y = u_y t - 0.5gt^2 \implies d \sin(10) = 24 \sin(55) t - 0.5(9.81)t^2$$

$$0.173d = 1.395d - 0.0247d^2 \implies d = 48.9m$$

2.3.3 Normal and Tangential Components:

1. In the n-t coordinate system, the origin is located on the particle (thus the origin and coordinate system move with the particle).

2. The t-axis is tangent to the path (curve) at the instant considered, and the n-axis is perpendicular to the t-axis.
3. The center of curvature, O', always lies on the concave side of the curve. The radius of curvature, ρ , is defined as the perpendicular distance from the curve to the center of curvature at that point.
4. The velocity vector is **always** tangent to the path of motion (t-direction).
5. **Velocity:** $\vec{v} = v\hat{u}_t$
6. **Acceleration:** $\vec{a} = v'\hat{u}_t + (v^2/\rho)\hat{u}_n$
7. **When Particle moves in a Straight Line:**
 $\rho \rightarrow \infty \implies a_n = 0 \implies a = a_t = v'$
8. **When Particle moves along a Curve at Constant Speed:**
 $a_t = v' = 0 \implies a = a_n = v^2/\rho$
9. **Radius of Curvature:**

$$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{d^2y/dx^2}$$

10. **Question:** A boat is traveling along a circular path with a speed of $v = (0.0625t^2)m/s$, where t is in seconds. Determine the magnitude of its velocity and acceleration when t = 10 s. [Given, $\rho = 40m$]

Answer:

$$v = (0.0625)(10)^2 = 6.25m/s$$

$$a_t = v' = (0.0625)(2t) = (0.125)(10) = 1.25m/s^2$$

$$a_n = \frac{v^2}{\rho} = \frac{(6.25)^2}{40} = 0.9766m/s^2$$

Hence,

$$a = \sqrt{a_t^2 + a_n^2} = 1.59m/s^2$$

11. **Question:** Starting from rest, a bicyclist travels around a horizontal circular path, r = 10 m, at a speed of $v = (0.09t^2 + 0.1t)m/s$, where t is in seconds. Determine the magnitudes of his velocity and acceleration when he has traveled s = 3 m.

Answer:

$$s = \int v dt = \int (0.09t^2 + 0.1t) dt = 0.03t^3 + 0.05t^2$$

$$\text{When } s = 3m, 3 = 0.03t^3 + 0.05t^2 \implies t = 4.147s$$

$$\text{At } t = 4.147s, v = 0.09(4.147)^2 + 0.1(4.147) = 1.96m/s$$

$$a_t = v' = 0.18t + 0.1 = 0.18(4.147) + 0.1 = 0.8465m/s^2$$

$$a_n = \frac{v^2}{\rho} = \frac{(1.96)^2}{10} = 0.3852m/s^2$$

$$a = \sqrt{a_t^2 + a_n^2} = 0.930m/s^2$$

12. The magnitude of the normal acceleration is inversely proportional to radius of curvature.
13. The directions of the tangential acceleration and velocity are always collinear.

2.3.4 Cylindrical Components:

1. We can express the location of P in polar coordinates as $\vec{r} = r\hat{u}_r$. Note that the radial direction, r , extends outward from the fixed origin, O, and the transverse coordinate, θ , is measured anti-clockwise from the horizontal.

2. Velocity in 2-D:

$$\vec{v} = r'\hat{u}_r + r\theta'\hat{u}_\theta$$

3. Acceleration in 2-D:

$$\vec{a} = (r'' - r(\theta')^2)\hat{u}_r + (r\theta'' + 2r'\theta')\hat{u}_\theta$$

4. Velocity in 3-D:

$$\vec{v}_p = r'\hat{u}_r + r\theta'\hat{u}_\theta + z'\hat{u}_z$$

5. Acceleration in 3-D:

$$\vec{a}_p = (r'' - r(\theta')^2)\hat{u}_r + (r\theta'' + 2r'\theta')\hat{u}_\theta + z''\hat{u}_z$$

6. **Question:** A platform is rotating about the vertical axis such that at any instant its angular position is $\theta = (4t^{3/2})rad$, where t is in seconds. A ball rolls outward along a radial groove so that its position is $r = (0.1t^3)m$, where t is in seconds. Determine the magnitudes of the velocity and acceleration of the ball when $t = 1.5$ s.

Answer:

$$r = 0.1t^3 \implies r' = 0.3t^2 \implies r'' = 0.6t$$

$$\theta = 4t^{3/2} \implies \theta' = 6t^{1/2} \implies \theta'' = 3t^{-1/2}$$

$$v = \sqrt{(0.3t^2)^2 + ((0.1t^3)(6t^{1/2}))^2} = \sqrt{(0.09(1.5)^4) + (0.6(1.5)^{7/2})^2}$$

Hence, $v = 2.57m/s$.

$$a_r = r'' - r(\theta')^2 = (0.6(1.5)) - (0.1(1.5)^3(36(1.5))) = -17.33m/s^2$$

$$a_\theta = r\theta'' + 2r'\theta' = (0.1(1.5)^3(3(1.5)^{-1/2})) + (2(0.3(1.5)^2)(6(1.5)^{1/2})) = 10.75m/s^2$$

Hence,

$$a = \sqrt{(-17.33)^2 + (10.75)^2} = 20.4m/s^2$$

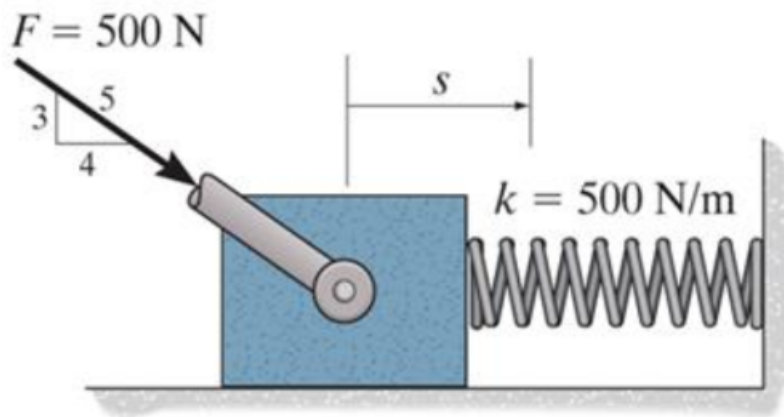
2.4 Newton's Laws of Motion:

1. A particle originally at rest, or in uniform motion, will remain in this state if the resultant force acting on the particle is zero.
2. If the resultant force on the particle is not zero, the particle experiences an acceleration in the same direction as the resultant force. This acceleration has a magnitude proportional to the resultant force.
3. Mutual forces of action and reaction between two particles are equal, opposite, and collinear.

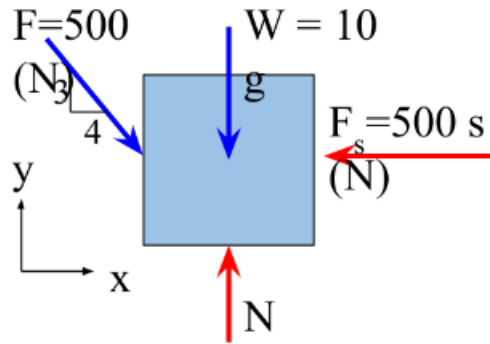
Some Key Points:

1. Newton's second law can be written in mathematical form as $\sum \mathbf{F} = m\mathbf{a}$. Within the summation of forces, $\sum \mathbf{F}$, internal forces are(is) not included.
2. Newton's second law is a "law of nature" - experimentally proven, not the result of an analytical proof.
3. Mass (a property of an object) is a measure of the resistance to a change in velocity of the object.
4. Weight (a force) depends on the local gravitational field. Calculating the weight of an object is an application of $\mathbf{F} = m\mathbf{a}$, i.e., $\mathbf{W} = m\mathbf{g}$.

Question: The spring is placed between the wall and the 10-kg block. If the block is subjected to a force of $F = 500\text{ N}$, Draw the free-body and kinetic diagrams of the block. When $s = 0$, the block is at rest and the spring is uncompressed. The contact surface is smooth.



Answer:

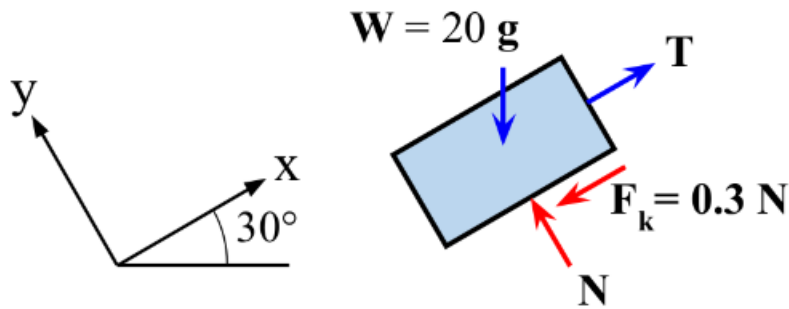


2.4.1 Rectangular Coordinates:

1. In dynamics, the friction force acting on a moving object is always kinetic.
2. If a particle is connected to a spring, the elastic spring force is expressed by $F = ks$. The “s” in this equation is the difference between deformed length and un-deformed length.
3. **Question:** A motor winds in a cable with a constant acceleration such that a 20-kg crate moves a distance $s = 6\text{m}$ in 3s , starting from rest. Determine the tension developed in the cable. The coefficient of kinetic friction between the crate and the plane is $\mu_k = 0.3$. The plane is inclined to the horizontal at an angle of 30° .

Solution:

$$\text{We know } s = ut + 0.5at^2 \implies 6 = 0 + 0.5a(3)^2 \implies a = 1.33\text{m/s}^2$$

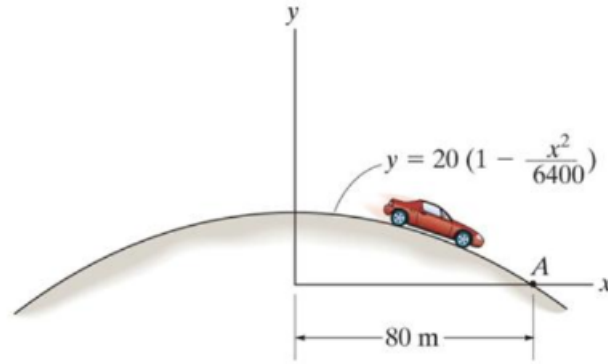


$$\Sigma F_y = N = mg\cos(30^\circ) = 0 \implies N = 169.9\text{N}$$

$$\Sigma F_x = T - 0.3N - mg\sin(30^\circ) = ma \implies T = 176\text{N}$$

2.4.2 Normal-Tangential Coordinates:

1. The positive n direction of the normal and tangential coordinates is always directed toward the center of curvature and is normal to the tangential & bi-normal components.
2. **Question:** The 800 kg car travels over the hill having the shape of a parabola. When the car is at point A, it is traveling at 9 m/s and increasing its speed at 3 m/s². Determine both the resultant normal force and the resultant frictional force that all the wheels of the car exert on the road at this instant. Neglect the size of the car.



Solution:

$$\Sigma F_n = mg \cos \theta - N = mv^2 / \rho$$

$$(800)(9.81) \cos \theta - N = (800)(3)^2 / \rho \implies N = 7848 \cos \theta - 64800 / \rho$$

$$\Sigma F_t = mg \sin \theta - f = ma_t$$

$$(800)(9.81) \sin \theta - f = (800)(3) \implies f = 7848 \sin \theta - 2400$$

$$\text{Now, } y = 20(1 - x^2/6400) \implies y' = -x/160 \implies y'' = -1/160$$

$$\rho \Big|_{x=80m} = \frac{[1 + (dy/dx)^2]^{3/2}}{d^2y/dx^2} = \frac{[1 + (-0.5)^2]^{3/2}}{0.00625} = 223.6m$$

Also,

$$\tan \theta = \left| \frac{dy}{dx} \Big|_{x=80m} \right| = |-0.5| \implies \theta = 26.6^\circ$$

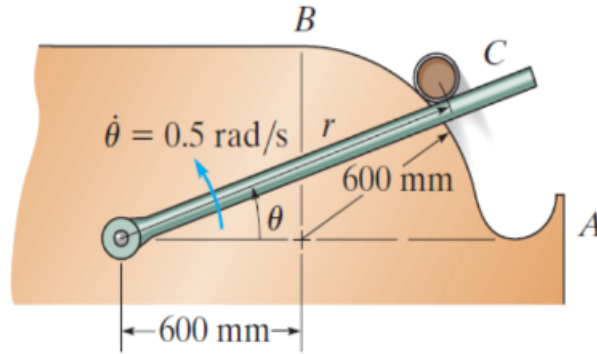
Hence,

$$N = 7848 \cos \theta - 64800 / \rho = 6728N$$

$$f = 7848 \sin \theta - 2400 = 1114N$$

2.4.3 Cylindrical Coordinates:

1. When the forces acting on a particle are resolved into cylindrical components, friction forces always act in the tangential direction.
2. **Question:** A smooth can C, having a mass of 3 kg, is lifted from a feed at A to a ramp at B by a rotating rod. If the rod maintains a constant angular velocity of $\dot{\theta} = 0.5 \text{ rad/s}$, determine the force which the rod exerts on the can at the instant $\theta = 30^\circ$. Neglect the effects of friction in the calculation and the size of the can so that $r = (1.2 \cos \theta) \text{ m}$. The ramp from A to B is circular, having a radius of 600 mm.



Solution:

$$r = 1.2 \cos \theta \implies r' = (-1.2 \sin \theta) \theta' \implies r'' = (-1.2 \cos \theta) \theta'^2 - (1.2 \sin \theta) \theta''$$

$$\theta = 30^\circ \implies \theta' = 0.5 \implies \theta'' = 0$$

$$\implies r = 1.039 \text{ m} \implies r' = -0.3 \text{ m/s} \implies r'' = -0.2598 \text{ m/s}^2$$

Now,

$$a_r = r'' - r \theta'^2 = -0.5196 \text{ m/s}^2$$

$$a_\theta = r \theta'' + 2r' \theta' = -0.3 \text{ m/s}^2$$

Now,

$$\Sigma F_r = N \cos \theta - mg \sin \theta = ma_r \implies N = 15.2 \text{ N}$$

$$\Sigma F_\theta = f + N \sin \theta - mg \cos \theta = ma_\theta \implies f = 17 \text{ N}$$

2.5 Work and Energy:

1. Work is defined as the product of force and displacement components acting in the same direction.
2. A force does work on a particle when the particle undergoes a displacement along the line of action of the force.

3.

$$dU = Fds\cos\theta \implies U = \int_{r_1}^{r_2} F \cdot dr$$

4. If the force and the displacement directions are perpendicular, the work is zero.

5. Work due to Weight:

$$U_W = \int_h mgy = mgh$$

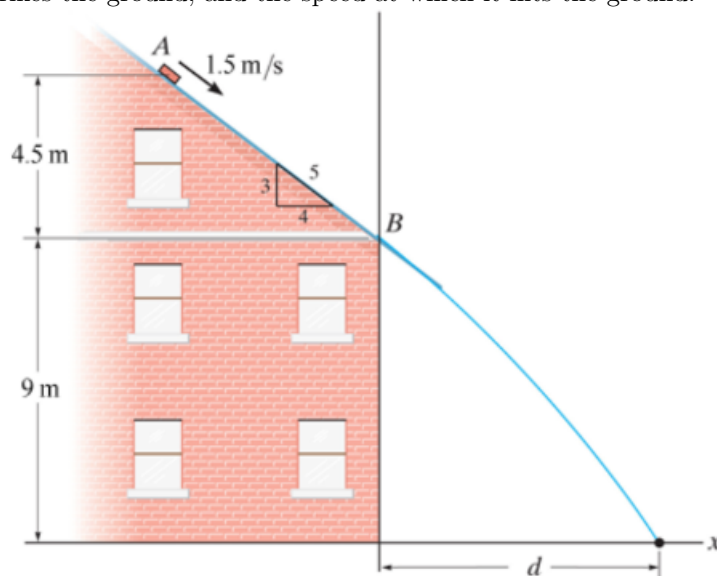
6. Work due to Spring:

$$U_s = \int F_s ds = \int_x kx ds = \frac{1}{2}kx^2$$

7. A particle's initial kinetic energy plus the work done by all the forces acting on the particle as it moves from its initial to final position is equal to the particle's final kinetic energy.

$$KE_i + U = KE_f$$

8. **Question:** The 1-kg brick slides down a smooth roof, such that when it is at A it has a velocity of 1.5 m/s. Determine the speed of the brick just before it leaves the surface at B, the distance d from the wall to where it strikes the ground, and the speed at which it hits the ground.



Solution: We know $KE_A + U = KE_B$

$$\frac{1}{2}mv_A^2 + mgh = \frac{1}{2}mv_B^2 \implies v_B = 9.515 \text{ m/s}$$

Now, brick performs projectile motion:

$$x_{BC} = v_{Bx}t \implies d = 7.612t$$

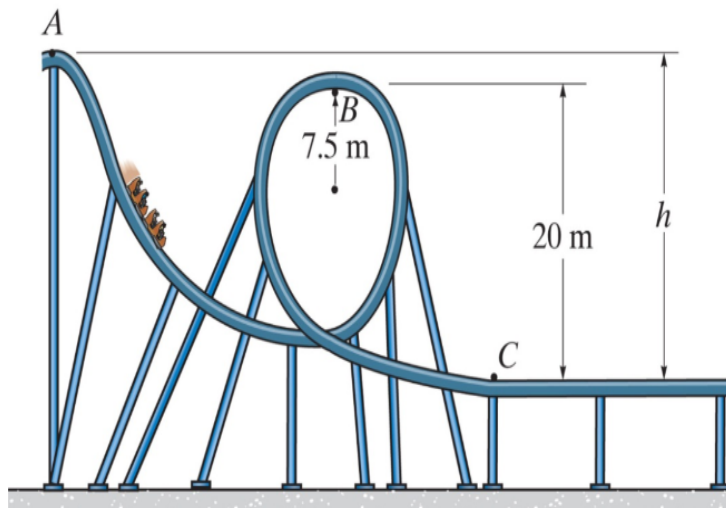
$$y_{BC} = v_{By}t - \frac{1}{2}gt^2 \implies t = 0.8923s$$

Hence, $d = 6.79m$. Also,

$$KE_A + U = KE_C \implies \frac{1}{2}mv_A^2 + mgH = \frac{1}{2}mv_C^2 \implies v_C = 16.34m/s$$

2.6 Conservation of Energy:

1. The potential energy of a spring is always positive.
2. A force F is said to be conservative if the work done is independent of the path followed by the force acting on a particle as it moves from A to B. This also means that the work done by the force F in a closed path (i.e., from A to B and then back to A) is zero.
3. Potential energy is a measure of the amount of work a conservative force will do when a body changes position.
4. **Law of Conservation of Mechanical Energy:** When a particle is acted upon by a system of conservative forces, the work done by these forces is conserved and the sum of kinetic energy and potential energy remains constant.
5. **Question:** The roller coaster car having a mass 800 kg is released from rest at point A. If the track is to be designed so that the car does not leave it at B, determine the required height h . Also, find the speed of the car when it reaches point C. Neglect friction.



Solution: By Conservation of Mechanical Energy between A and B,

$$\frac{1}{2}mv_A^2 + mg(h - 20) = \frac{1}{2}mv_B^2$$

At point B,

$$N = 0 \implies mg = mv_B^2/\rho \implies v_B^2 = 8.578m/s$$

$$\therefore 0 + mg(h - 20) = 0.5(8.578)^2 \implies h = 23.75m$$

By Conservation of Mechanical Energy between A and C,

$$\frac{1}{2}mv_A^2 + mg(h - 20) = \frac{1}{2}mv_C^2 \implies v_C = 21.6m/s$$

2.7 Linear Momentum - Impulse Principle:

1. The linear momentum (p) and impulse (J) equation is obtained by integrating the equation of motion with respect to time.

$$J = \Delta p \implies \int F dt = m\Delta v$$

2. **Question:** The 30-Mg freight car A ($v=20\text{km/hr}$) and 15-Mg freight car B ($v=10\text{km/hr}$) are moving towards each other with the velocities shown. Determine the maximum compression of the spring ($k=3\text{MN/m}$) mounted on car A. Neglect rolling resistance.

Solution: By COLM, $m_A v_A + m_B v_B = (m_A + m_B)v$

$$(30)(20) + (15)(10) = (45)v \implies v = 16.667\text{km/hr}$$

By COME, $KE_i + U_i = KE_f + U_f$

$$[0.5(30)(20)^2 + 0.5(15)(10)^2] + 0 = 0.5(45)(16.667)^2 + 0.5(3 \times 10^6)x^2$$

$$12000 + 1500 = 6250.25 + 1500000x^2 \implies x = 481\text{mm}$$

2.8 Impact:

1. Impact occurs when two bodies collide during a very short time period, causing large impulsive forces to be exerted between the bodies.
2. Central impact occurs when the directions of motion of the two colliding particles are along the line of impact.
3. Oblique impact occurs when the direction of motion of one or both of the particles is at an angle to the line of impact.
4. The coefficient of restitution, e , is the ratio of the particles' relative separation velocity after impact to the particles' relative approach velocity before impact. It is an indicator of the energy lost during the impact.

5. **Question:** A girl throws the ball with a horizontal velocity $v_1 = 2.4 \text{ m/s}$ from a height of 0.9m . If the coefficient of restitution between the ball and the ground is 0.8 , determine (a) the velocity of the ball just after it rebounds from the ground and (b) the maximum height to which the ball rises after the first bounce.

Solution: We know, $v_y^2 = u_y^2 - 2gh$ and $u_y = 0$ as it is thrown horizontally.

$$v_y^2 = 2(9.81)(-0.9) \implies v_y = 4.202 \text{ m/s}$$

$$e = \frac{v_{2y}}{v_y} \implies 0.8 = \frac{v_{2y}}{4.202} \implies v_{2y} = 3.362 \text{ m/s}$$

Also, as $v_x = u_x = v_{2x} \implies v_{2x} = 2.4 \text{ m/s}$ Now,

$$v_{2y}^2 = 2gh \implies (3.362)^2 = 2(9.81)h \implies h = 0.576 \text{ m}$$

2.9 Angular Momentum:

- The angular momentum of a particle about point O is defined as the “moment” of the particle’s linear momentum about O.
-

$$\vec{L}_O = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v} \implies L_O = mvr$$

- The resultant moment acting on the particle about point O is equal to the time rate of change of the particle’s angular momentum.
- Principle of Angular Momentum and Impulse:**

$$\tau = \frac{dL}{dt} \implies \int_{t_1}^{t_2} \tau dt = L_2 - L_1$$

- Conservation of Angular Momentum:** When the sum of angular impulses acting on a particle or a system of particles is zero during the time t_1 to t_2 , the angular momentum is conserved.
- Question:** Each ball has a negligible size and a mass of 10 kg and is attached to the end of a rod whose mass may be neglected. If the rod is subjected to a torque $M = (t^2 + 2) \text{ N.m}$, where t is in seconds, determine the speed of each ball when $t = 3 \text{ s}$. Each ball has a speed $v = 2 \text{ m/s}$ when $t = 0$. Each ball is at a distance of 0.5m from the rod axis.

Solution:

$$\tau = t^2 + 2 \implies \int_0^3 (t^2 + 2) dt = \left[\frac{1}{3}t^3 + 2t \right]_0^3 = 9 + 6 = 15 \text{ N.m.s}$$

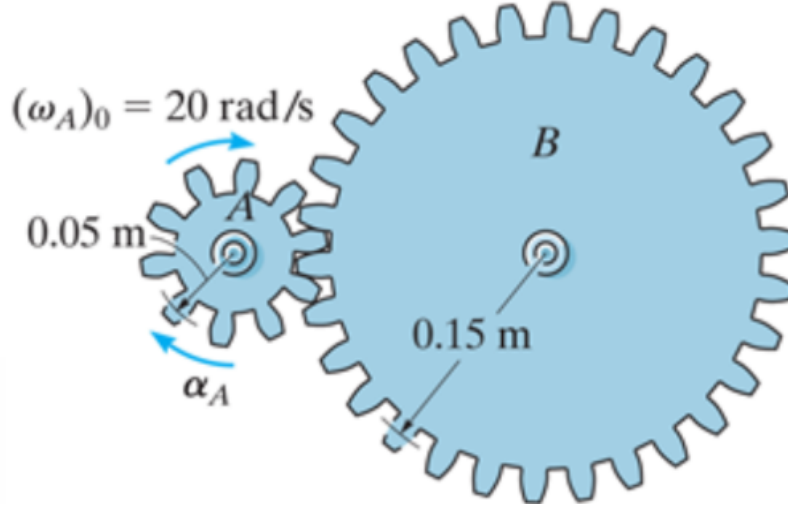
$$\Delta L = L_2 - L_1 = 2((10)(2)(0.5)) - 2((10)v(0.5)) = 10(2 - v)$$

$$\therefore \int \tau dt = \Delta L = 15 = 10(2 - v) \implies v = 3.5 \text{ m/s}$$

- If a particle moves in the $x - y$ plane, its angular momentum vector is in the z - direction.

2.10 Planar Rigid Body Motion:

1. If a rigid body is in translation only, the velocity at points A and B on the rigid body are always the same.
2. For two gears A (radius r_A) and B (radius r_B) in contact at a point, $\theta_A r_A = \theta_B r_B$, $\omega_A r_A = \omega_B r_B$, and $\alpha_A r_A = \alpha_B r_B$.
3. **Question:** A motor gives gear A an angular acceleration of $\alpha_A = 4t^3 \text{ rad/s}^2$, where t is in seconds. If this gear is initially turning at $(\omega_A)_0 = 20 \text{ rad/s}$, determine the angular velocity and position of gear B when $t = 2 \text{ s}$.



$$\omega_A = (\omega_A)_0 + \int_0^t \alpha_A dt = 20 + \int_0^t 4t^3 dt = t^4 + 20$$

$$\theta_A = (\theta_A)_0 + \int_0^t \omega_A dt = 0 + \int_0^t (t^4 + 20) dt = \frac{1}{5}t^5 + 20t$$

For $t = 2 \text{ s}$, $\omega_A = 36 \text{ rad/s}$ and $\theta_A = 46.4 \text{ rad}$

Now, we know that

$$\omega_A r_A = \omega_B r_B \implies \omega_B = 36(0.05/0.15) = 12 \text{ rad/s}$$

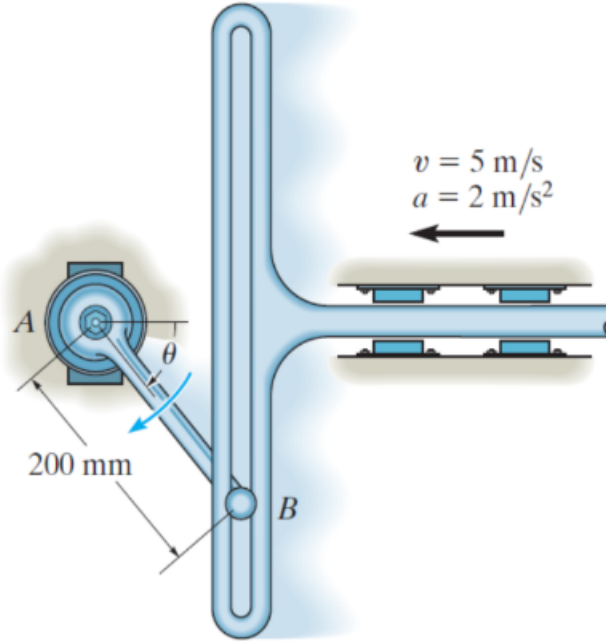
Similarly,

$$\theta_A r_A = \theta_B r_B \implies \theta_B = 46.4(0.05/0.15) = 15.46 \text{ rad}$$

2.11 Absolute Motion Analysis:

1. A body subjected to general plane motion undergoes a/an simultaneous translation and rotation.
2. In general plane motion, if the rigid body is represented by a slab, the slab rotates about an axis perpendicular to the plane.

3. **Question:** At the instant $\theta = 60^\circ$, the slotted guide rod is moving to the left with an acceleration of 2 m/s^2 and a velocity of 5 m/s . Determine the angular acceleration and angular velocity of link AB at this instant.



By looking at the figure, $x = 0.2 \cos \theta$ m.

$$\therefore v = -0.2 \sin \theta \times \omega \implies -5 = -0.2(0.866)\omega \implies \omega = 28.9 \text{ rad/s}$$

$$\begin{aligned} \therefore a &= -0.2 \cos \theta \times \omega^2 - 0.2 \sin \theta \times \alpha \\ \implies -2 &= -0.2(0.5)(28.9)^2 - 0.2(0.866)\alpha \implies \alpha = 470 \text{ rad/s}^2 \end{aligned}$$